# ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

## Chapter 13: Statistics (Sankhyiki)

### Key Concepts & Important Formulas:

* Mean of Grouped Data:
1. Direct Method: Mean = Sigma(fi * xi) / Sigma(fi)
2. Assumed Mean Method: Mean = a + Sigma(fi * di) / Sigma(fi), where di = xi - a

* Mode of Grouped Data:
Mode = l + [(f1 - f0) / (2*f1 - f0 - f2)] * h, where l is lower limit of modal class.

* Median of Grouped Data:
Median = l + [(n/2 - cf) / f] * h, where cf is cumulative frequency of class preceding median class.

* Empirical Relationship between Measures of Central Tendency:
3 * Median = Mode + 2 * Mean

### Solved Exam Examples & Practice:

Example: If mean = 20 and median = 22, find mode using empirical relationship.
Mode = 3*Median - 2*Mean = 3(22) - 2(20) = 66 - 40 = 26.